Nifty 50 Stock Market Prediction: A Machine Learning Approach

Predicting HDFCBANK Closing Prices

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Objective: Predicting Tomorrow's Stock Price

- **Goal:** To build and evaluate regression models that predict the next day's closing price of a stock (HDFCBANK) using the current day's trading data.
- **Problem Type:** This is framed as a supervised machine learning regression task.
- **Key Question:** Can we accurately forecast the closing price using features like 'Open', 'High', 'Low', and 'Turnover'?

Our Technical Approach

1. Data Preparation

- **Data Loading & Cleaning:**
  - Source: `HDFCBANK.csv`
  - Cleaned infinite values and dropped nulls.
- **Feature Engineering:**
  - Features (X): Open, High, Low, Last, Turnover
  - Target (y): Next Day's Close Price
- **Data Splitting:**
  - Chronological 80/20 Train-Test split.

2. Modeling & Evaluation

- **Model Training:**
  - Trained five different regression models.
- **Hyperparameter Tuning:**
  - Used GridSearchCV for SVR and KNN models.
- **Evaluation & Comparison:**
  - Assessed models using R^2, MAE, and RMSE metrics to find the best performer.

Models & Hyperparameter Tuning

Models Implemented

- Multiple Linear Regression
- Bayesian Ridge Regression
- Random Forest Regressor
- Support Vector Regression (SVR)
- K-Nearest Neighbors (KNN) Regressor

Note on Scaling: Features were scaled using `StandardScaler` for SVR and KNN.

Hyperparameter Tuning

Method: GridSearchCV

SVR Tuning:

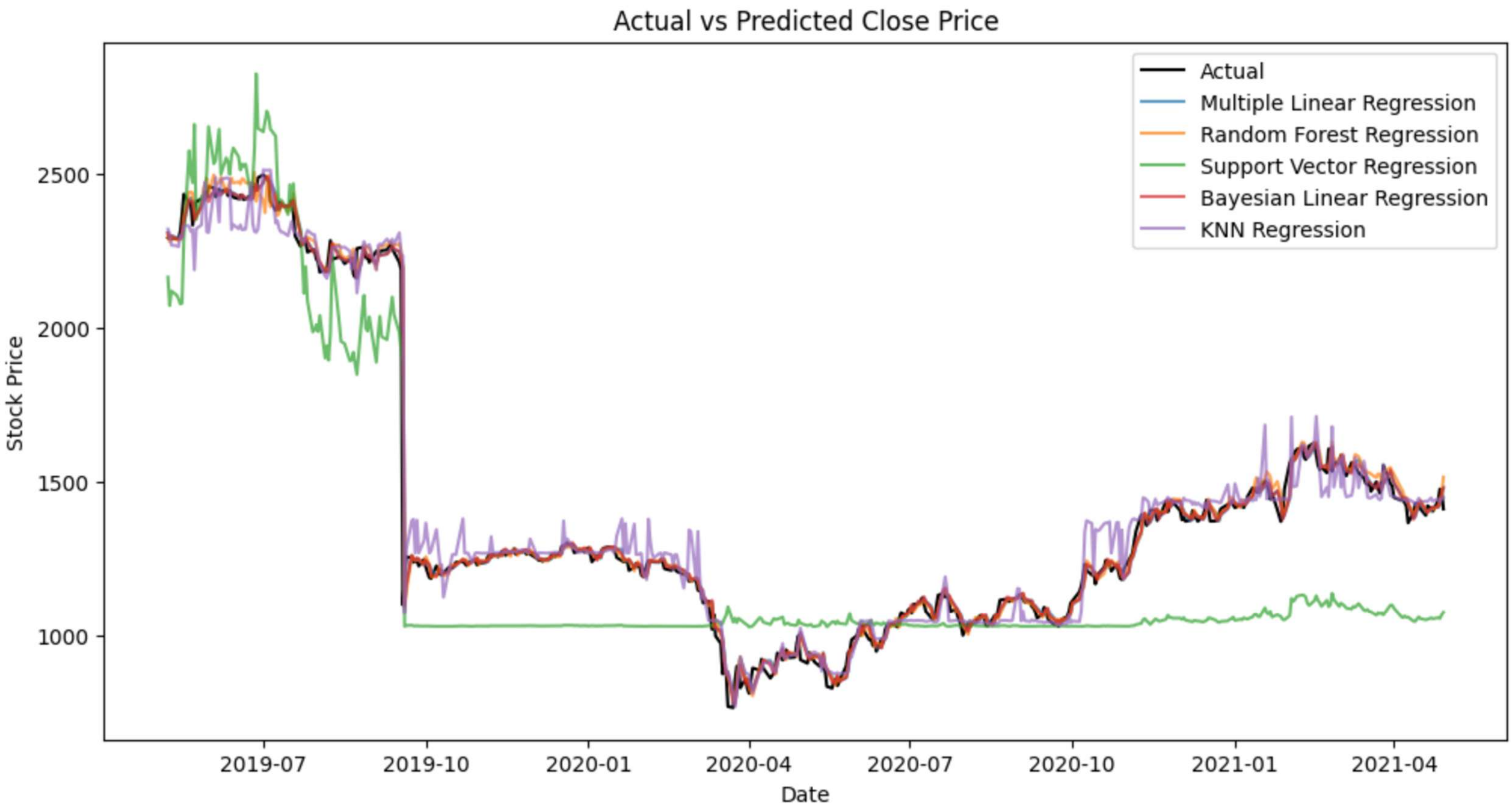
- Kernels: `poly`
- Best Params: C=0.1, degree=3, epsilon=1

KNN Tuning:

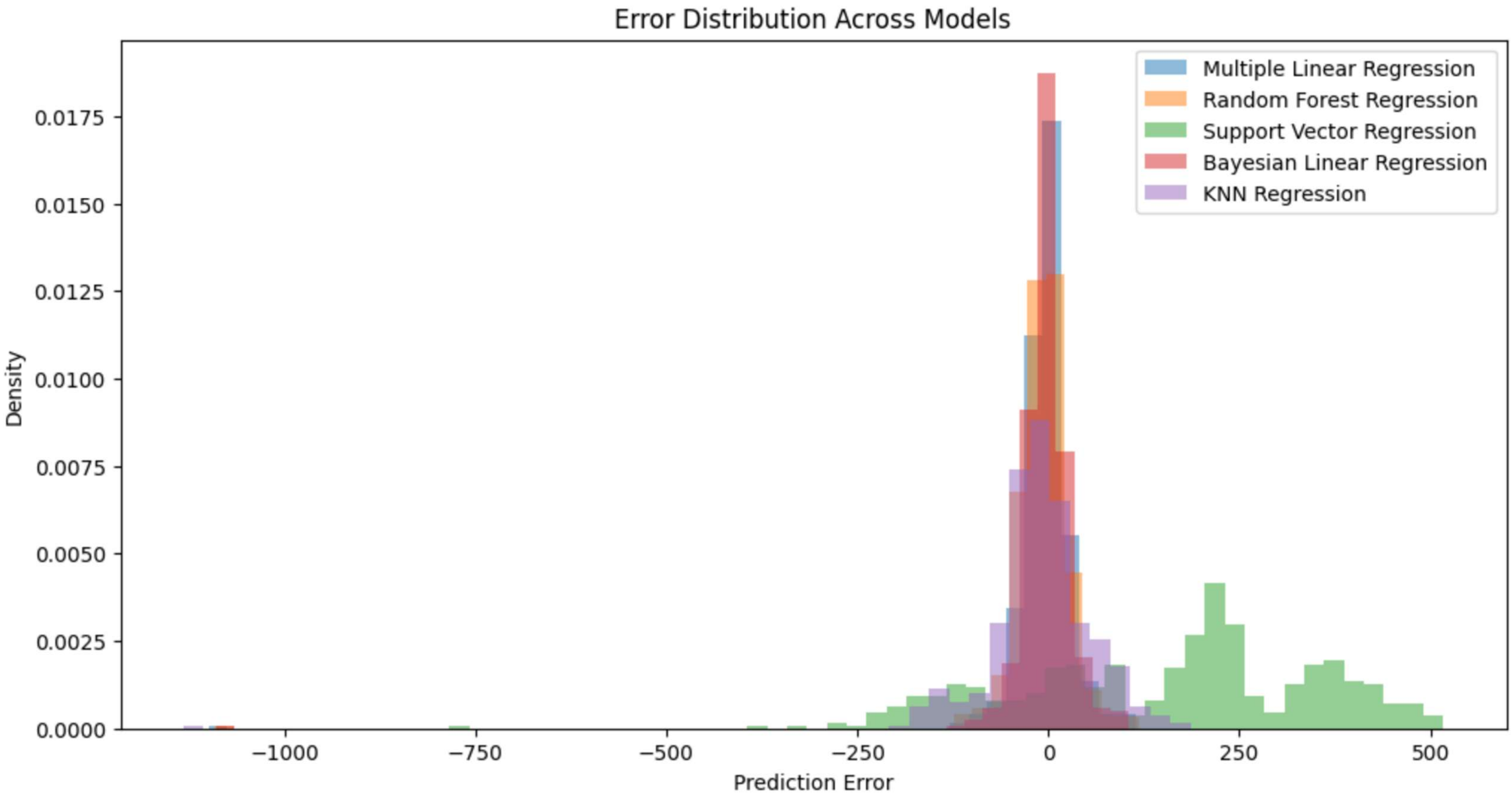
- Metrics: Manhattan, Euclidean
- Best Params: n_neighbors=3, p=1, weights='distance'

Model Performance on Test Data

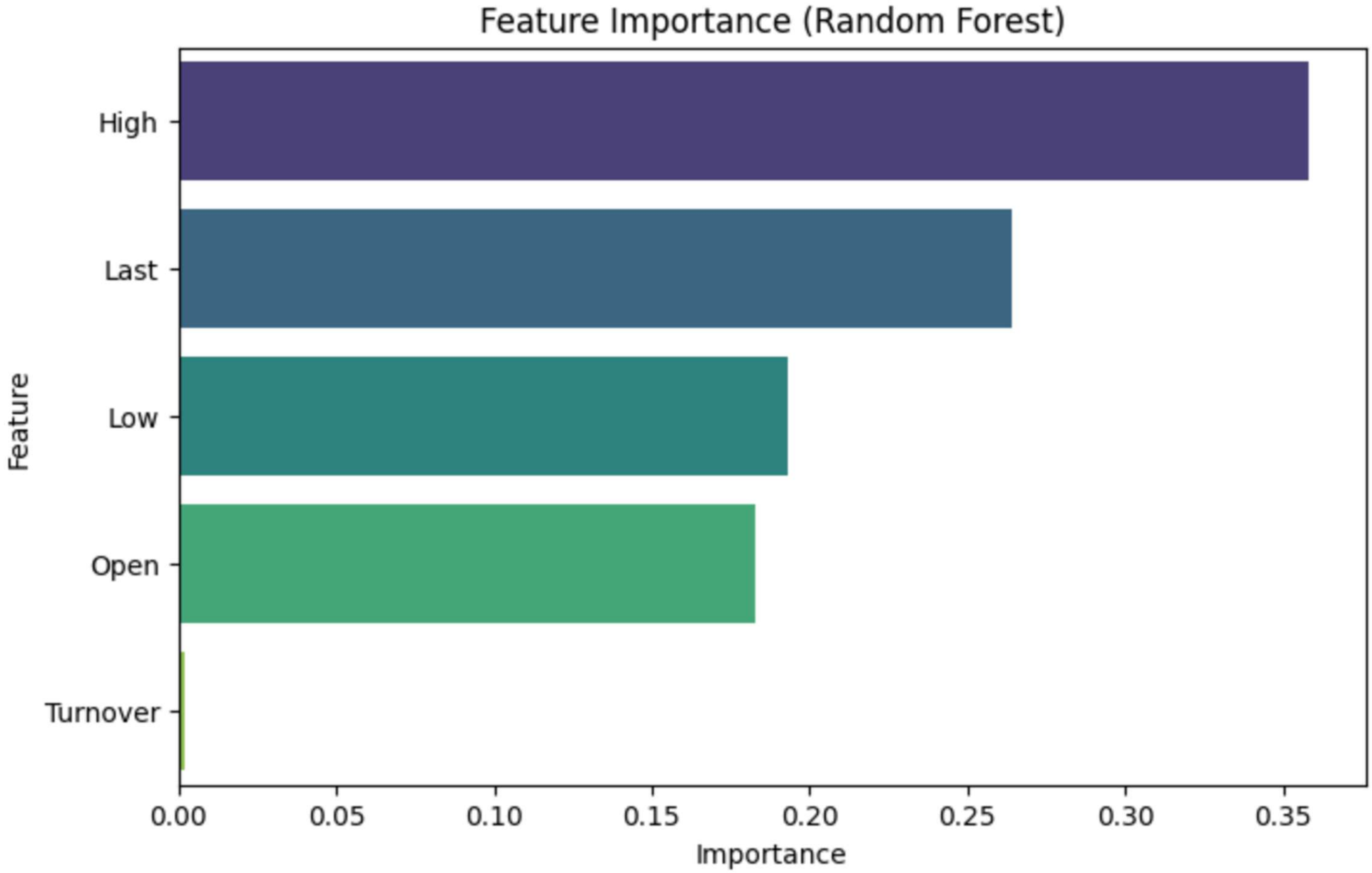
Model	MAE	RMSE	R^2 Score
Bayesian Linear Regression	**21.82**	**56.22**	**0.9855**
Multiple Linear Regression	22.66	57.30	0.9850
Random Forest Regression	25.42	58.60	0.9843
KNN Regression	47.60	79.29	0.9712
Support Vector Regression	207.91	246.53	0.7215

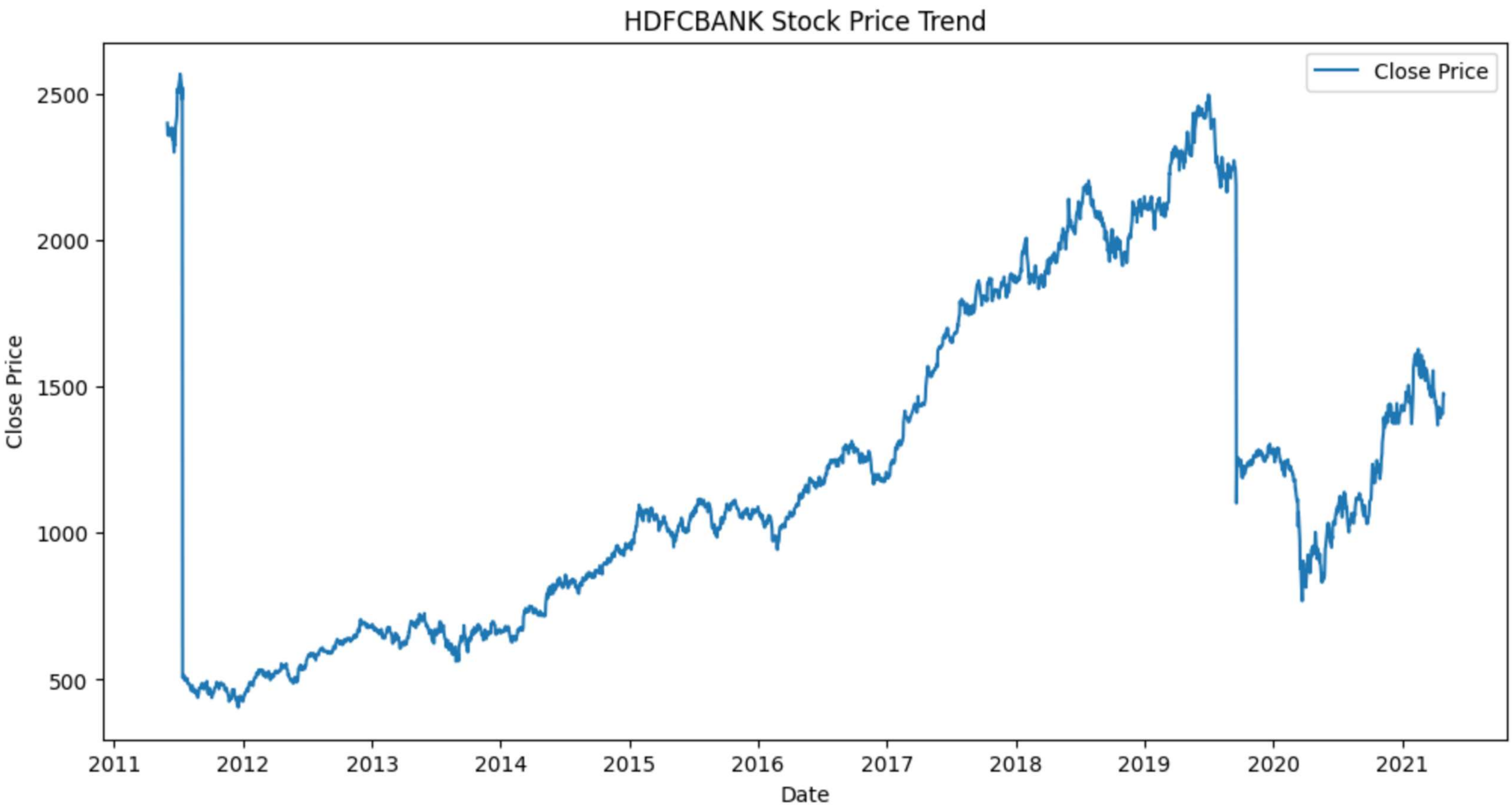


Actual vs Predicted Close Price
Actual
Multiple Linear Regression
Random Forest Regression
Support Vector Regression
Bayesian Linear Regression
KNN Regression
2500
2000
1500
1000
Stock Price
2019-07
2019-10
2020-01
2020-04
2020-07
2020-10
2021-01
2021-04
Date



Error Distribution Across Models
Multiple Linear Regression
Random Forest Regression
Support Vector Regression
Bayesian Linear Regression
KNN Regression
Density
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
−1000
−750
−500
−250
0
250
500
Prediction Error


Feature Importance (Random Forest)
High
Last
Low
Open
Turnover
Feature
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Importance



HDFCBANK Stock Price Trend
Close Price
2500
2000
1500
1000
500
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
Date
Close Price

Conclusion & Next Steps

Conclusion

- Daily trading data is a strong predictor for the next day's closing price.
- Bayesian Linear Regression proved to be the most effective model for this dataset.
- Turnover and Last price were identified as the most important features.

Future Work

- Incorporate more features like moving averages (MA), RSI, or Bollinger Bands.
- Experiment with time-series models like LSTMs or ARIMA.
- Integrate sentiment analysis from news headlines as an additional feature.